

MINGZE JIANG

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EDUCATION

Tsinghua University

Aug 2023-Jun 2027(expected)

B.E. in Computer Science and Technology

GPA: Overall GPA of 3.9/4.0 **Rank:** 7/172 (top 5%)

GRE 339 (Quantitative 170 / Verbal 169) + 3.5 (AW)

TOEFL 113 (Listening 30) || IELTS 8.0 (Listening 9.0 / Reading 9.0)

RELEVANT COURSEWORK: Introduction to Artificial Intelligence (A+), Introduction to Complex Analysis (A+), Software Engineering (A+), Fundamentals of Programming (A), Data Structures and Algorithms (A), Digital Image Processing (A), Programming and Training (Python) (A), Fundamentals of Computer Graphics (A), Digital Logic Experimentation (A), Physics for Scientists and Engineers (A), Calculus (A), Discrete Mathematics (A), Probability and Statistics (A)

PUBLICATIONS

- **Mingze Jiang**, Xueyang Li, John Kheir, Alec Gerten, Yiyu Shi. “*LightRefine-PCXR: A Lightweight Refinement Framework for Efficient Medical Device Suppression in Pediatric Chest X-Rays*”, accepted by Int. Conf. Medical Imaging with Deep Learning (MIDL), 2026.
- Xueyang Li, **Mingze Jiang**, Gelei Xu, Jun Xia, Ching-Hao Chiu, Mengzhao Jia, Danny Chen, Yiyu Shi. “*CRC-Router: Risk-Constrained Routing for Medical Agentic AI Systems*”, submitted to Int. Conf. Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2026.

RESEARCH EXPERIENCES

Mori Laboratory, Nagoya University, Japan Visiting Student, Advisor: Prof. Kensaku Mori *Jan 2024-Feb 2024*

▪ Medical Image Processing Fundamentals and Endoscopic Super-resolution Imaging

- Investigated resolution limitations of endoscopic imaging due to small sensor sizes.
- Developed a multi-frame registration and fusion method to enhance image clarity and sharpness, improving endoscopic detail.

Knowledge Engineering Group, Tsinghua University Group Member, Advisor: Prof. Yuxiao Dong *Mar 2024-Jul 2025*

▪ Multimodal Large Models for Visual Understanding & Generation

- Worked on large-scale multimodal AI models for visual understanding and generation.
- Contributed to dataset construction and model fine-tuning for multimodal tasks in the CogAgent project, focusing on vision-language models.

iSURE Program, University of Notre Dame, USA Research Intern, Advisor: Prof. Yiyu Shi *Jul 2025-present*
on-site Jul 2025-Aug 2025, remote since Sep 2025

▪ AI Framework for Pediatric Critical Care: Automated CXR Analysis & Anomaly Detection

- Worked with real pediatric ICU chest X-ray data to develop an AI framework for clinically meaningful quantification and decision support (*Collaboration with Boston Children's Hospital/Harvard Medical School*).
- Proposed LightRefine-PCXR, a lightweight, anatomy-preserving framework for medical device suppression in pediatric chest X-rays that combines coarse inpainting with localized refinement to achieve high-fidelity restoration at low training cost.
- Proposed a topology-aware deformable centerline-based segmentation method for tubular devices in pediatric chest X-rays, targeting continuity-preserving multi-class predictions under overlap and occlusion.
- Contributed to the development of an agentic AI-based triage system for chest X-ray assessments, integrating uncertainty-aware decision-making to optimize clinical workflows and support reliable decision support.

SCHOLARSHIP

2025 **Comprehensive Excellence Scholarship**, Tsinghua University

2024 **Comprehensive Excellence Scholarship**, Tsinghua University

2023 **Freshman Scholarship**, Tsinghua University

HONORS & AWARDS

2022 **National Gold** in 36th Annual AAPT (**American** Association of Physics Teachers) Physics Bowl Contest
2022 **Global Gold** in **British** Physics Olympiad Senior Physics Challenge
2022 **Global Silver** in **British** Physics Olympiad Competition (BPHOR2)
2022 **National Silver, Global Top 150** in **Canada** SIN (Sir Isaac Newton) Contest
2022 **Canadian Top 20, China National Silver** in CAP (**Canadian** Association of Physicists) High School Physics Prize
2021 **Top Gold** in **British** Physics Olympiad Competition (BPHO R1)

SERVICE & LEADERSHIP

Field Research on Technology Equity and Digital Divide

CORE MEMBER

Jul 2024-Aug 2024

- Conducted literature research on regional historical context and industrial policies, synthesizing findings into a preliminary research report of over 10,000 words.
- Investigated regional development and industrial upgrading, analyzing pathways to deploy emerging technologies (e.g. big data, IoT, AI) in inland areas with limited geographical advantages.
- Published research outcomes in news media to raise public awareness and advocated for regional computing resource development and digital technologies implementation.
- Awarded the Tsinghua University Student Social Practice Silver Award.

Association of Student Academic Development of Tsinghua University

CORE MEMBER

Mar 2025-present

- Directed the Promotion Team for Tsinghua's "Buddy Program", a peer-mentorship initiative that builds learning communities through structured online/offline engagement, helping students establish consistent academic routines.
- Organized the Reading Initiative and Graduation Check-in Campaign, produced and published multiple official account articles achieving over 1,000 views.

Inspire Epistolary Mentorship Program of Tsinghua University

Remote

VOLUNTEER

Feb 2025-Jul 2025

- Built cross-regional mentorship relationships with secondary school students in under-resourced regions via long-term handwritten letters, fostering personal connection and perspective sharing.

Yinhe Correspondence Program of Tsinghua University

Remote

VOLUNTEER

Sep 2024-Jan 2025

- Participated in a special education volunteer initiative, exchanging handwritten letters with hearing-impaired students at Beijing Qiyin Experimental School to foster social connection, provide mentorship, and promote mutual understanding and emotional support.